## 2. Explain the four values of the Agile Manifesto with suitable examples.

**Answer :**

The Agile Manifesto gives four main values that guide Agile software development. These values help teams work better, deliver faster, and satisfy customer needs.

**Four Values of Agile Manifesto :**

1) Individuals and interactions over processes and tools

Explanation : Agile gives more importance to people, teamwork, and communication than only following rules or depending only on tools.

Example : A team solves a problem quickly by discussing it together instead of waiting for a formal report.

2) Working software over comprehensive documentation

Explanation : Agile gives more importance to delivering software that works than writing too many documents.

Example : A team shows a working login page to the customer instead of only giving long design documents.

3) Customer collaboration over contract negotiation

Explanation : Agile encourages regular communication with the customer so the product matches their needs.

Example : The customer gives feedback after every sprint, and the team updates the product accordingly.

4) Responding to change over following a plan

Explanation : Agile accepts changes even during development if they improve the product.

Example : A shopping app team adds a new payment option after the customer requests it during development.

In short, the Agile Manifesto values focus on people, working software, customer involvement, and flexibility to change. These values help in building better software.

## 4. What are Agile Methods? Explain the classification of Agile methods.

**Answer :**

Agile Methods are a set of software development approaches based on Agile values and principles. These methods focus on iterative development, customer collaboration, flexibility and delivering working software in short cycles.

- **Agile Methods :**

  Agile methods are lightweight processes that help teams respond quickly to changes and deliver value to customers through continuous improvement and feedback.

- **Classification of Agile Methods :**

  Agile methods can be classified in the following ways. (Any one classification is given below.)

**1. Based on Approach**

- Iterative and Incremental Methods : Software is developed in small increments with repeated cycles.
  Example : Scrum, XP
- Adaptive Methods : These methods focus on continuous adaptation to change.
  Example : DSDM, Crystal
- Lean Methods : These methods aim to maximize value and minimize waste.
  Example : Kanban, Lean Software Development

**2. Based on Team Organization**

- Team-based Methods : Small, self-organizing teams do the planning and development.
  Example : Scrum, XP
- Collaborative Methods : Emphasis on close collaboration between developers and customers.
  Example : Crystal, FDD

5. Define Agile Project Management.

Explain its major characteristics and benefits.

**Answer :**

Agile Project Management is a way of managing projects that focuses on delivering work in small parts called iterations. It follows Agile values and principles to deliver useful software quickly and adapt to changes easily.

## Major Characteristics of Agile Project Management :

① Iterative and Incremental

- The project is divided into small parts (iterations). Each iteration produces a working part of the product.

② Customer Collaboration

- Customers are involved throughout the project. Their feedback helps in making the right product.

③ Flexibility and Adaptability

- Agile accepts changes in requirements even late in development. The plan can be adjusted as needed.

④ Frequent Communication

- Regular communication among team members, and with customers helps in solving problems quickly and keeps everyone aligned.

⑤ Short Delivery Cycles

- Working software is delivered in short cycles (weeks). This helps in early value delivery and continuous improvement.

## Benefits of Agile Project Management :

- Delivers working software early and continuously.
- Easily adapts to changes in requirements.
- Improves customer satisfaction through regular feedback.
- Reduces risks by solving issues early.
- Increases transparency and visibility of project progress.
- Encourages teamwork and improves productivity.

In short, Agile Project Management helps teams deliver the right product to the customer, at the right time, with better quality and less risk.

6. Compare the Traditional Model and Agile Model with respect to requirements, planning, development, and customer involvement.

**Answer :**

Traditional Model (Waterfall) and Agile Model are different in the way they handle requirements, planning, development and customer involvement.

| Aspect | Traditional Model (Waterfall) | Agile Model |
|---|---|---|
| Requirements | • All requirements are gathered at the beginning.<br>• They are fixed and rarely changed.<br>• Focus is on complete documentation. | • Requirements are gathered continuously.<br>• Changes are expected and welcomed.<br>• Focus is on working software and customer value. |
| Planning | • Planning is done in detail before starting the project.<br>• The whole project plan is created upfront.<br>• Hard to change the plan later. | • Planning is done in short cycles (iterations or sprints).<br>• Plans are flexible and can change.<br>• Focus is on short term goals. |
| Development | • Development happens in phases (one after another).<br>• Coding starts after design is fully completed.<br>• Working software is delivered only at the end. | • Development happens in small iterations.<br>• Design, coding, testing happen together.<br>• Working software is delivered frequently. |
| Customer Involvement | • Customer is involved mainly at the beginning and end.<br>• Less feedback during the development.<br>• Customer sees the product only at the end. | • Customer is involved throughout the project.<br>• Regular feedback and reviews are encouraged.<br>• Customer sees working product frequently. |

**In Short :**

Traditional Model is plan-driven and follows a fixed process.
Agile Model is change-driven and focuses on flexibility, collaboration and delivering value in small steps.

## 7. Explain Agile Team Interactions and discuss their importance in successful Agile projects.

**Answer :**

Agile Team Interactions means how team members communicate, collaborate and work together throughout the project. In Agile, people and their interactions are more important than processes and tools.

### Key Agile Team Interactions :

① Daily Stand-up Meetings : Team members meet for a short time every day to share what they did yesterday, what they will do today and any problems.

② Continuous Communication : Team members talk to each other regularly and share ideas, doubts and feedback.

③ Collaboration : Developers, testers, designers and business members work together to solve problems and make decisions.

④ Feedback and Reviews : The team shows their work to the customer or other members and takes feedback to improve it.

⑤ Retrospectives : At the end of each iteration, the team discusses what went well, what can be improved and how to do better next time.

### Importance of Agile Team Interactions :

- Improves understanding among team members.
- Helps in solving problems quickly.
- Builds trust and teamwork.
- Increases transparency in work.
- Leads to better quality software.
- Helps in delivering the right product that meets customer needs.

**In Short :**

Good interaction in Agile teams leads to better communication, faster problem solving, happy customers and successful projects.

8. Explain the importance of Ethics in Agile Teams with suitable examples.

Answer :

Ethics in Agile teams means following good values and right behaviour while working together. It includes honesty, respect, responsibility, fairness and trust. Ethics is important in Agile because team members work closely, communicate often and make quick decisions.

Importance of Ethics in Agile Teams :

1) Builds trust : Team members trust each other when everyone is honest and responsible.
2) Improves teamwork : Respectful behaviour helps developers, testers and managers work together smoothly.
3) Encourages open communication : Ethical teams speak truthfully about progress, problems and mistakes.
4) Supports customer satisfaction : Teams act responsibly and deliver what is promised to the customer.
5) Reduces conflicts : Fairness and respect help avoid arguments and misunderstandings.
6) Improves quality : When team members act sincerely, they test properly and do not hide defects.
7) Creates a healthy work culture : Ethical behaviour makes the workplace positive, safe and professional.

Suitable Examples :

1) Honest reporting : During a daily stand-up, a developer clearly says that a task is delayed instead of hiding the problem. This helps the team plan correctly.
2) Respect for team members : In a sprint meeting, everyone listens to each other's ideas politely, even when opinions are different.
3) Responsibility towards quality : A tester finds a bug and reports it properly instead of ignoring it just to finish work early.
4) Fair dealing with customer : If a feature cannot be completed in the sprint, the team informs the customer honestly and gives a realistic update.

In Short :

Ethics in Agile teams is important because it builds trust, improves teamwork, supports quality and helps deliver better software in a fair and responsible way.